Engineering Report – AeroGrid Real-Time Turbine Monitoring System

From the data processing results, it is clear that turbines 4 and 7 are failing based on elevated temperature and vibration levels identified by the anomaly detection script. For the cloud architecture, firstly, Apache Kafka is used as the message streaming layer, acting as a buffer between turbine sensors and processing systems. This allows turbines to send data continuously and is stored temporarily in a durable event stream. This design prevents data loss during system failures or traffic spikes. AWS Lambda is used for stream processing, running the anomaly detection logic in real time. It is serverless and automatically scales with incoming data volume, ensuring efficient processing without the need for dedicated servers.

For storage, multiple layers are used, Amazon S3 Raw storage, ensuring raw data is preserved as a backup before processing. Amazon Timestream is used as hot storage to store recent turbine data, optimised for fast querying in the dashboards. Amazon S3 Data Lake is used as cold storage for historical data to be stored cheaply for long-term analysis and even later ML applications.

This architecture improves cost efficiency, by separating hot and cold storage, data is automatically transition between the storage tiers using S3 lifecycle policies, ensuring only frequently access data remains in expensive storage, while older data is moved to lower-cost archival storage but is still accessible.

Finally, on the User-side, the dashboard uses Amazon QuickSight to visualise turbine health metrics, connecting directly with Timestream and allowing real-time monitoring of turbine performance and failure detection. An Amazon SNS-based alerting system sends real-time notifications when turbine anomalies are detected by AWS Lambda, enabling immediate response via email, SMS or monitoring dashboard to prevent system failures,